

# VeinXam: A Low-Cost Deep Veins Function Assessment

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## ABSTRACT

A venous insufficiency, to which the deep veins of the lower human extremities are particularly susceptible, can lead to serious diseases, such as a deep vein thrombosis (DVT) with subsequent risks of severe implications, e.g. pulmonary embolism or a post-thrombotic syndrome (PTS) [6]. The current standard procedure to diagnose venous insufficiency is performed exclusively in medical offices and hospitals in the form of in-patient treatments with special medical equipment. This hurdle for the patient, combined with the often diffuse symptoms of venous insufficiency [7], may lead to a late discovery of diseases such as DVTs and increases the risk of secondary diseases as well as treatment costs [3]. To address these issues, we propose a novel method for continuous monitoring of the current venous function by adapting the Light Reflection Rheography (LLR) and using low-cost wearable sensor technology and a smartphone app, aiming to deliver critical early stage information about pathological changes of the blood flow in the lower limbs.

## CCS CONCEPTS

• **Human-centered computing** → **Mobile devices**.

## KEYWORDS

Deep Vein Thrombosis; Light Reflection Rheography; Wearables

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## SYSTEM OVERVIEW AND RESULTS

As before mentioned, our solution adapts the LRR, a non-invasive diagnostic tool which is used to simply and quickly evaluate the function of lower limb blood vessels via blood flow data acquired from a photoplethysmogram (PPG) [1, 5] and can thus be employed to develop a wearable aiming to prevent venous diseases of lower limbs. To achieve this, we miniaturised the hardware necessary to perform an LRR analysis: The body-worn sensor unit used to record the PPG consists of an optical analogue front end, an LED, and a photodiode designed for 880 nm infrared light, a skin temperature sensor, a battery, and a system-on-chip, which transmits the recorded data via Bluetooth Low Energy. Due to its small dimensions of 40×23×10 mm, the unit can be attached to the calf either with medical tape or with a compression stocking without interfering with everyday life. The recorded PPG is processed on a custom developed smartphone app, which includes all necessary data processing steps to identify the LRR signal and main parameters, the venous refill time  $T_0$  and the Venous Pumping Capacity  $V_0$ . Additionally, an animated guide within the app informs new users about the correct seating position and the LRR typical venous muscle pump manoeuvre. Finally,  $T_0$  and  $V_0$  are compared to clinically established thresholds [2, 4] and visualised using a simple colour-coding system with textual descriptions to warn the user in case of detected insufficiencies.

To evaluate our proposed solution, 5 data tracks of an individual with a diagnosed PTS on one and a healthy venous vascular bed in the other leg have been recorded. For medical reasons, the subject had to wear a compression stocking on the leg with PTS, which gave us the possibility to test both intended mounting options for the developed sensor unit. The subject's medical track record fortunately included two LRR examinations taken after its diagnosed DVT and PTS. Calculating the mean over each estimated  $T_0^i$  for the 5 data tracks of both the healthy and the leg diagnosed with PTS yielded  $mean(T_{0_{healthy}}^i) = 31.83 s$  and  $mean(T_{0_{PTS}}^i) = 14.5 s$  which are close to the  $T_{0_{healthy}} = 25 s$  and the  $T_{0_{PTS}} = 10 s$  of the subject's medical record. When interpreted using the insufficiency grades of [2, 4], our solution provides an equal medical assessment compared to the professional results of the subject's past medical examinations, suggesting the validity of our proposed wearable deep veins function assessment solution.

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